

SANJAY KUMAR S

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Professional Summary

Master of Computer Applications (MCA) graduate and software developer with hands-on experience in full-stack web development, cybersecurity, and machine learning. Proficient in building scalable applications using Python, Django, MERN Stack, REST APIs, JavaScript, SQL, and modern web technologies, with practical experience gained through research internships and end-to-end software projects. Strong foundation in secure software development, problem-solving, and collaborative engineering, with a proven ability to design, develop, and deliver innovative technology solutions.

Skills

Programming: Python, C++, Java, JavaScript, TypeScript, SQL, HTML, CSS, Visual Basic

Frameworks & Tools: Django, MERN Stack, Git, GitHub, Flask, UiPath Studio

Technical Skills: REST APIs, CRUD Operations, API Integration, Authentication, Responsive Web Design, Machine Learning, Cybersecurity

Databases: SQLite, MySQL, MongoDB,

Soft Skills: Problem Solving, Leadership, Team Collaboration, Adaptability, Time Management

Experience

SwiftSafe

March 2026 – May 2026

Cybersecurity Machine Learning & AI Researcher Intern

- Developed Python-based cybersecurity solutions for threat detection using static analysis, automated alerting, and AI-driven security workflows.
- Built secure applications for remote access control, centralized logging, and real-time monitoring following secure coding practices.
- Researched cybersecurity threats, vulnerabilities, and attack patterns to develop AI-driven security frameworks that supported secure coding practices and proactive threat mitigation.

Skyslit Network Private Limited

2024

- Contributed to web application development workflows by supporting application architecture analysis, development lifecycle activities, and deployment processes.
- Collaborated with the documentation and development teams to prepare technical documentation, review project workflows, and support software development activities.
- Assisted in project coordination, documentation management, and requirement analysis, ensuring accurate technical documentation and streamlined team collaboration.

Projects

SOUNDWAVE: Speech Assistance for People with Stuttering

Tech Stack: Python, ML, Dart, Flask

- Developed an AI-powered speech assistance system to improve communication fluency for people with stuttering by integrating speech recognition, next-word prediction, and real-time speech synthesis into a unified workflow.
- Top-3 next-word prediction accuracy of 85.4%, and an average end-to-end response latency of approximately **370 ms**.

CloudVault – Secure AWS Cloud File Management System

Tech Stack: AWS S3, SQS, Lambda, DynamoDB, Cognito, Flask, React, Python

- Built a secure cloud-based file management system with Cognito authentication, private S3 storage, and a user-scoped Flask REST API.
- Implemented asynchronous S3 → SQS → Lambda processing with DynamoDB metadata, secure pre-signed downloads, and file status tracking.

Cybersecurity Monitoring and Incident Response System

Tech Stack: SIEM, Anomaly Detection, Python , Flask

- Designed a comprehensive logging and real-time alerting system.
- Implemented automated anomaly detection and centralized event monitoring to support faster incident identification and response.

Emotion-Based Music System

Tech Stack: Python, ML, OpenCV, Django

- Developed an ML-powered system that detects user emotion and recommends music.
- Integrated emotion recognition and personalized recommendation techniques to enhance user experience and adaptive music selection.

Delicious Food Recipe Web Application

Tech Stack: Django, SQLite, HTML, CSS, JavaScript, REST API

- Developed a full-stack Django recipe management application supporting CRUD operations and API-driven calorie estimation, improving nutritional information accessibility through responsive web interfaces.

Plant-Based E-Commerce Web Application

Tech Stack: MERN Stack, JWT, MongoDB

- Developed a full-stack e-commerce platform with RESTful workflows, role-based authentication, and modular backend services supporting product, user, and order management.
- Developed RESTful workflows for product management, user access control, and seamless shopping experiences.

Publications

Necrobotics: The Integration of Biotic Materials as Ready-to-Use Actuators and Chassis

IJARES, ISSN No: 2455-6211

Paper ID: IJ-2702261005

Sustainable robotics approach to repurpose biological remains involving biodegradable actuators and bio-inspired engineering systems.

Education

Master of Computer Applications

2024 – 2026

Nehru College of Engineering and Research Centre, Pampady, Thiruvilwamala
APJ Abdul Kalam Technological University(KTU)

CGPA: 9.59

Bachelor of Computer Applications

2021 – 2024

College of Applied Science, Perissery Kerala University
CGPA: 7.89

Senior Secondary Education

2019 – 2021

Amrita Vidyalayam, Thiruvalla
Central Board of Secondary Education

Certifications

Microsoft Azure AI Essentials
Full Stack Development with Python & Django
Advanced Gemini API for Integration
IHRD Tharang: Data Mining Workshop

NPTEL: Database Management System
IBM SkillBuild AI Fundamentals
Advanced SQL for Performance Optimization
K-DISC Young Innovator Program

Volunteer Experience

National Service Scheme (NSS)

2021 – 2023

- Coordinated volunteer activities, awareness drives, and meetings.
- Participated in National Integration Camp and State-Level NSS Camp.
- Led awareness drives and social service activities